



- 1 The times taken by members of a large quiz club to complete a challenge have a normal distribution with mean μ minutes. The times, x minutes, are recorded for a random sample of 8 members of the club. The results are summarised as follows, where $\bar{x}$ is the sample mean.

$$\bar{x} = 33.8 \quad \sum (x - \bar{x})^2 = 94.5$$

Find a 95% confidence interval for μ .

[4]

We know that C.I. = $\bar{x} \pm t_{(1-\frac{\alpha}{2}), n-1} \cdot \frac{s}{\sqrt{n}}$

$$n=8, \\ s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{94.5}{7} = 13.5$$

$$t_{(1-\frac{\alpha}{2}), n-1} = t_{0.975, 7} = 2.365$$

So,

$$C.I = 33.8 \pm 2.365 \times \sqrt{\frac{13.5}{8}} = 33.8 \pm 3.072$$

$$= (30.7, 36.9) \text{ minutes}$$

- 2 A scientist is investigating the size of shells at various beach locations. She selects four beach locations and takes a random sample of shells from each of these beaches. She classifies each shell as large or small. Her results are summarised in the following table.

		Beach location				Total
		A C ₁	B C ₂	C C ₃	D C ₄	
Size of shell	Large R ₁	68	69	96	81	314
	Small R ₂	28	55	64	39	186
Total		96	124	160	120	500

Test, at the 10% significance level, whether the size of shell is independent of the beach location. [7]

H₀: Shell size is independent of beach location.

H₁: Shell size is not independent of beach location.

$$\text{Expected frequency } E_{ij} = \frac{R_i \times C_j}{T}, T=500$$

O _{ij}	E _{ij}	O _{ij} ² /E _{ij}
68	60.288	76.6985
69	77.872	61.1388
96	100.48	91.7197
81	75.36	87.0621
28	35.712	21.9534
55	46.128	65.5784
64	59.52	68.8172
39	44.64	34.0726

$\sum O_{ij}^2/E_{ij} - N$ method for test statistic is not explicitly in Mark Scheme, but you can use this method.

$$\text{Test statistic } \chi^2 = \sum \left(\frac{O_{ij}^2}{E_{ij}} \right) - N = 76.6985 + 61.1388 + \dots + 34.0726 - 500 = 7.0407 \approx 7.041$$

$$v = (r-1)(c-1) = (2-1)(4-1) = 3$$

$$\text{Critical value } \chi^2_{3}(0.10) = 6.251$$

7.041 > 6.251, so we reject H₀. There is sufficient evidence to suggest that shell size is not independent of beach location.

- 3 George throws two coins, A and B , at the same time. Coin A is biased so that the probability of obtaining a head is a . Coin B is biased so that the probability of obtaining a head is b , where $b < a$. The probability generating function of X , the number of heads obtained by George, is $G_X(t)$. The coefficients of t and t^2 in $G_X(t)$ are $\frac{5}{12}$ and $\frac{1}{12}$ respectively.

(a) Find the value of a .

[2]

Coefficient of t^2 means both heads on A and B , so

$$ab = \frac{1}{12} \rightarrow b = \frac{1}{12a} \quad (1)$$

Probability of 0 heads is $1 - \frac{5}{12} - \frac{1}{12} = \frac{1}{2}$, so

$$(1-a)(1-b) = \frac{1}{2} \quad (2)$$

Substitute (1) in (2),

$$(1-a)(12a-1) = 6a,$$

$$12a - 12a^2 - 1 + a = 6a,$$

$$12a^2 - 7a + 1 = 0, \quad a = \frac{7 \pm 1}{24} = \frac{1}{3}, \frac{1}{4}. \text{ Since our equations were symmetric, one of these is } b. \text{ Since } b < a, \\ a = \frac{1}{3}.$$

The random variable Y is the sum of two independent observations of X .

(b) Find the probability generating function of Y , giving your answer as a polynomial in t .

[3]

$$G_X(t) = \frac{1}{2} + \frac{5}{12}t + \frac{1}{12}t^2$$

$$G_Y(t) = G_{X_1+X_2}(t) = G_{X_1}(t) \cdot G_{X_2}(t) = [G_X(t)]^2, \text{ therefore}$$

$$G_Y(t) = \left(\frac{1}{12}\right)^2 \cdot (6 + 5t + t^2)^2 = \frac{1}{144} (36 + 25t^2 + t^4 + 60t + 12t^2 + 10t^3)$$

$$= \frac{1}{144} (36 + 60t + 37t^2 + 10t^3 + t^4)$$

(c) Find $\text{Var}(Y)$.

[3]

$$G'_Y(t) = \frac{1}{144} \cdot (60 + 74t + 30t^2 + 4t^3)$$

$$G'_Y(1) = \frac{7}{6}$$

$$G''_Y(t) = \frac{1}{144} \cdot (74 + 60t + 12t^2)$$

$$G''_Y(1) = 73/72$$

$$\text{Var}(Y) = G''_Y(1) + G'_Y(1) - [G'_Y(1)]^2$$

$$= 73/72 + 7/6 - 49/36$$

$$\text{Var}(Y) = \frac{59}{72} = 0.819 \text{ (3 sf)}$$

- 4 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{3}{8} \left(1 + \frac{1}{x^2}\right) & 1 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find $E(\sqrt{X})$.

[3]

$$E(\sqrt{X}) = \int_1^3 \left(\frac{3}{8} \sqrt{x} + \frac{3}{8x^{3/2}} \right) dx$$

$$= \left[\frac{x^{3/2}}{4} - \frac{3}{4\sqrt{x}} \right]_1^3 = \frac{3\sqrt{3}}{4} - \frac{\sqrt{3}}{4} + \frac{1}{2}$$

$$E(\sqrt{X}) = \frac{1}{2}(\sqrt{3}+1)$$

The random variable Y is given by $Y = X^2$.

- (b) Find the probability density function of Y .

[4]

CDF of X , t

$$F(t) = \int_1^t \left(\frac{3}{8} + \frac{3}{8x^2} \right) dx = \left[\frac{3}{8}x - \frac{3}{8x} \right]_1^t = \frac{3}{8}t - \frac{3}{8t}$$

CDF of Y $G(y) = P(Y \leq y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = F(\sqrt{y})$

So,

$$G(y) = \frac{3}{8}\sqrt{y} - \frac{3}{8\sqrt{y}} : 1 \leq \sqrt{y} \leq 3$$

PDF of Y $g(y) = G'(y)$, so

$$g(y) = \begin{cases} \frac{3}{16\sqrt{y}} + \frac{3}{16y^{3/2}} & 1 \leq y \leq 9 \\ 0 & \text{otherwise} \end{cases}$$

(c) Find the 40th percentile of Y .

[3]

$$G(a) = 0.4,$$

$$\frac{3}{8} \left(\sqrt{a} - \frac{1}{\sqrt{a}} \right) = 0.4, \quad \left(\sqrt{a} - \frac{1}{\sqrt{a}} \right)^2 = \left(\frac{16}{15} \right)^2,$$

$$a + \frac{1}{a} - 2 = \frac{256}{225},$$

$$\frac{a^2 + 1}{a} = \frac{706}{225}, \quad a^2 - \frac{706}{225}a + 1 = 0$$

$$a = \frac{25}{9}, \frac{9}{25} \text{ but only } \frac{25}{9} \text{ in domain, so}$$

$$40^{\text{th}} \text{ percentile is } \frac{25}{9}.$$

- 5 A manager claims that the lengths of the rubber tubes that his company produces have a median of 5.50 cm. The lengths, in cm, of a random sample of 11 tubes produced by this company are as follows.

5.56 5.45 5.47 5.58 5.54 5.52 5.60 5.35 5.59 5.51 5.62

It is required to test at the 10% significance level the null hypothesis that the population median length is 5.50 cm against the alternative hypothesis that the population median length is not equal to 5.50 cm.

Show that both a sign test and a Wilcoxon signed-rank test give the same conclusion and state this conclusion. [9]

$H_0: m = 5.50$, $H_1: m \neq 5.50$ where m is the population median length

Sign test:

Lengths	5.56	5.45	5.47	5.58	5.54	5.52	5.60	5.35	5.59	5.51	5.62
Sign of (lengths - m)	+	-	-	+	+	+	+	-	+	+	+

$S^+ = 8$, for $X \sim B(11, 0.5)$,

$$P(X \geq S^+) = P(X \geq 8) = ({}^{11}C_8 + {}^{11}C_9 + {}^{11}C_{10} + {}^{11}C_{11}) \times 0.5^{11}$$

$$= 0.113 \text{ (3 sf)}$$

Since two tailed, probability at upper tail = 0.05

$0.113 > 0.05$, so do not reject H_0 .

Signed-rank test:

Length	Length-m	P	N
5.56	0.06	6	
5.45	-0.05		5
5.47	-0.03		3
5.58	0.08	7	
5.54	0.04	4	
5.52	0.02	2	
5.60	0.10	9	
5.35	-0.15		11
5.59	0.09	8	
5.51	0.01	1	
5.62	0.12	10	
Σ		$P = 47$	$N = 19$

So, test statistic $T = \min(P, N) = \min(47, 19) = 19$

At two tailed $\alpha = 0.1, n = 11$, critical value $T^* = 13$

$19 > 13$, so do not reject H_0 .

Thus, both tests have the same conclusion.

There is insufficient evidence to suggest that the population median length is not equal to 5.50 cm.

- 6 A company has two machines, A and B , which independently fill small bottles with a liquid. The volumes of liquid per bottle, in suitable units, filled by machines A and B are denoted by x and y respectively. A scientist at the company takes a random sample of 40 bottles filled by machine A and a random sample of 50 bottles filled by machine B . The results are summarised as follows.

$$\Sigma x = 1120 \quad \Sigma x^2 = 31400 \quad \Sigma y = 1370 \quad \Sigma y^2 = 37600$$

The population means of the volumes of liquid in the bottles filled by machines A and B are denoted by μ_A and μ_B .

- (a) Test at the 2% significance level whether there is any difference between μ_A and μ_B . [8]

→ Samples are independent.

→ n_A and n_B are large, so it is a Z -test. ($n_A + n_B = 40 + 50 > 30$)

→ it is 2 tailed

$$H_0: \mu_A = \mu_B, H_1: \mu_A \neq \mu_B$$

$$\bar{x} = \frac{\Sigma x}{n_A} = \frac{1120}{40} = 28, \bar{y} = \frac{1370}{50} = 27.4$$

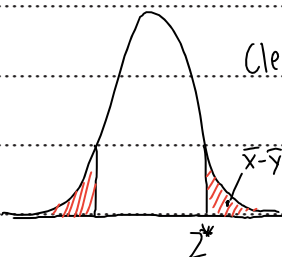
$$s_x^2 = \frac{1}{n_A - 1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n_A} \right) = \frac{1}{39} \left(31400 - \frac{1120^2}{40} \right) = \frac{40}{39}. \text{ Similarly,}$$

$$s_y^2 = \frac{62}{49}.$$

$$\text{Test statistic } Z = \frac{\bar{x} - \bar{y} - (\mu_A - \mu_B)}{\sqrt{\frac{s_x^2}{n_A} + \frac{s_y^2}{n_B}}} = \frac{28 - 27.4}{\sqrt{\frac{1}{39} + \frac{31}{1225}}} = 2.658$$

$$\text{Critical value } Z^* = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right) = \Phi^{-1}(0.99) = 2.326$$

Plot the sampling distribution of $\bar{A} - \bar{B}$,



Clearly, Z is in the rejection/critical region.

$2.658 > 2.326$, so we do reject H_0 .

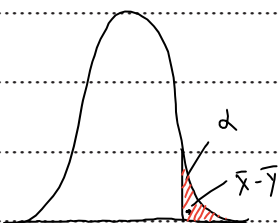
There is sufficient evidence to suggest that there is a difference between μ_A and μ_B .

- (b) Find the set of values of α for which there would be evidence at the $\alpha\%$ significance level that $\mu_A - \mu_B$ is greater than 0.25. [4]

$$H_0: \mu_A - \mu_B = 0.25, H_1: \mu_A - \mu_B > 0.25$$

$$Z = \frac{\bar{x} - \bar{y} - (\mu_A - \mu_B)}{\sqrt{\frac{s_x^2}{n_A} + \frac{s_y^2}{n_B}}} = \frac{0.6 - 0.25}{\sqrt{\frac{2434}{47775}}} = 1.550629726 \approx 1.551$$

Draw the sampling distribution,



we need all such α so $Z^* \leq Z$.

$$\text{So } Z^* \leq 1.551, \text{ p-value} \geq 1 - \Phi(1.551) = 1 - (0.9394 + 0.0001) = 0.0605$$

$\alpha\%$ is our required p-value, so $\alpha\% \geq 0.0605$.

Thus, the set of values of α is $\boxed{\alpha \geq 6.05}$

[illegible]

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